

ANKESH KUSHWAHA

Greater Noida, Uttar Pradesh, India

+91-7068122575 — ankeshkush9651@gmail.com

[LinkedIn](#) — [GitHub](#) — [Portfolio](#) — [LeetCode](#)

PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate focused on **Backend Development, Software Engineering, and Distributed Systems**. Built distributed backend systems using **Node.js, Express.js, Redis, MongoDB, PostgreSQL, WebSockets, and Docker**. Developed a distributed online coding platform with isolated code execution and Redis Streams-based worker architecture, load-tested with **800 concurrent submissions**. Solved **1,200+ DSA problems** with a **peak LeetCode rating of 1750**.

EDUCATION

Bachelor of Technology (Computer Science & Engineering)
G.L. Bajaj Institute of Technology & Management, Greater Noida

2023 – 2027
CGPA: **7.7**

TECHNICAL SKILLS

Languages: C++, Java, JavaScript, TypeScript, SQL

Frontend: HTML, CSS, Tailwind CSS, shadcn/ui, React.js

Backend: Node.js, Express.js, REST APIs, WebSockets, JWT, RBAC

Databases: PostgreSQL, MongoDB, Redis

Testing Framework: jest, vitest, cypress

Distributed Systems: Redis Streams, Redis Pub/Sub, Message Queues, Worker Architecture, Asynchronous Processing, Caching

DevOps & Tools: Docker, Docker Compose, Git, GitHub, CI/CD

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

PROJECTS

TalentIQ – Distributed Online Coding Platform

[GitHub Link](#)

React.js, Node.js, Express.js, Redis Streams, MongoDB, Docker, WebSockets, JWT, RBAC

- Built an online coding platform supporting **C++, Java, and Python** execution through isolated Docker containers.
- Designed a distributed **Redis Streams-based worker architecture** to asynchronously distribute code-execution tasks across horizontally scalable workers.
- Load-tested the code-execution pipeline with **20 concurrent virtual users**, processing 800+ submission requests with **0% errors** and approximately **923 ms average response time**.
- Implemented Docker-based execution isolation with **CPU, memory, and network restrictions** for securely executing untrusted user code.
- Implemented Redis caching for frequently accessed data, reducing measured API latency from approximately **100 ms to 10 ms**.
- Implemented **WebSocket-based real-time execution-status updates**, reducing reliance on repeated client polling.
- Secured backend APIs with **JWT authentication, RBAC, and rate limiting** to control access and resource usage.
- Integrated **LLM-powered debugging assistance** to provide contextual explanations for compilation and runtime errors.

Chatty – Real-Time Messaging Platform

[Live Link](#)

React.js, Node.js, Express.js, Redis Pub/Sub, MongoDB, Docker, WebSockets, JWT

- Built a real-time messaging platform using **WebSockets** for bidirectional communication and room-based messaging.
- Designed **Redis Pub/Sub** to enable real-time communication across horizontally scaled backend instances.
- Load-tested the system with **50 simulated users, 500 messages, and 4 backend instances** to validate real-time communication.
- Implemented **Redis caching**, reducing measured API latency from approximately **200 ms to 50 ms**.
- Designed the backend for **horizontal scaling and reliable cross-instance message delivery**.

ACHIEVEMENTS & OPEN SOURCE

- Solved **1,200+ Data Structures & Algorithms problems** across LeetCode, GeeksforGeeks, CodeStudio, and Codeforces.
- Achieved a **Peak LeetCode Rating of 1750**.
- Contributed to the **openSUSE** open-source ecosystem with **2 successfully merged pull requests**.
- **Smart India Hackathon:** Qualified for the finals and secured **3rd place among 400 teams** at the college level.